

DIVYAN ACHARYA

Mechanical Engineer | Postgraduate Student | University of Newcastle
0403 767 825 • divyanacharya777@gmail.com • Newcastle, NSW, Australia

PROFESSIONAL SUMMARY

Mechanical Engineering postgraduate student at the University of Newcastle with hands-on experience in mechanical maintenance, plant reliability, CAD modelling, FEA simulation, and technical fault diagnosis across industrial and aviation environments. Achieved a distinction-level GPA of 77.5% in Bachelor of Mechanical Engineering, and holds a positive Skills Assessment as a Professional Mechanical Engineer from Engineers Australia.

KEY STRENGTHS

- Design & Analysis:** Practical experience designing mechanical systems (motor-driven assemblies, shafts, bearings) and validating designs using FEA.
- Maintenance & Reliability:** Preventive maintenance programs, structured fault diagnosis, and asset documentation for industrial plant.
- Simulation-Driven Design:** Proficient in Ansys for static, dynamic, and structural analysis.
- Energy & Efficiency Focus:** Strong interest in reducing energy consumption and improving operational efficiency in industrial systems.
- Industry Experience:** Cross-sector exposure across industrial manufacturing and aviation maintenance, with safety-critical compliance experience.

EDUCATION

- Master of Professional Engineering (Mechanical)** 2026 – 2027
University of Newcastle — Callaghan Campus, NSW
- Focus areas: advanced mechanical systems, engineering management, and industry practice.
- Bachelor of Mechanical Engineering (Engineers Australia – Professional Engineer, positive skills assessment)** 2021 – 2025
Thapathali Campus, IOE, Tribhuvan University — Kathmandu, Nepal | GPA: 77.5% (Distinction)
- Consistently strong performance across structural mechanics, thermodynamics, manufacturing, and design.

PROFESSIONAL EXPERIENCE

- Graduate Mechanical Engineer** May 2025 – Oct 2025
Prasiddha Water Industries — Kalikanagar, Nepal
- Designed a motor-driven mechanical cleaning system (shaft, bearings, brush mechanism) — improving throughput and reducing manual handling risk.
 - Developed and implemented a preventive maintenance program for RO systems, filling machines, and packaging equipment, improving machine uptime.
 - Documented maintenance actions, fault histories, and corrective solutions, building a structured asset management knowledge base.
 - Applied SOPs for fault identification and resolution across plant operations.
- Mechanical Engineering Intern** Nov 2024 – Feb 2025
Nepal Airlines Corporation — Kathmandu, Nepal
- Assisted in maintenance and inspection of aircraft mechanical systems, ensuring compliance with aviation safety standards and CAAN regulations.
 - Applied structured fault diagnosis across aircraft systems, strengthening root cause analysis applicable to complex industrial environments.
 - Prepared technical documentation including maintenance records and compliance reports.
- CAE Engineer — Electric Vehicle Project** Feb 2023 – May 2024
Team Shireto, Thapathali Campus — Kathmandu, Nepal
- Conducted FEA on EV chassis and structural components using Ansys and HyperMesh (Optistruct), optimising designs for weight, stiffness, and safety.
 - Collaborated in a multidisciplinary team to integrate structural, mechanical, and electrical system architectures.

TECHNICAL SKILLS

CAD & Design: SolidWorks, Fusion 360, AutoCAD
FEA & Simulation: Ansys (static, dynamic, structural), Altair HyperMesh / Optistruct
Maintenance & Reliability: Preventive maintenance scheduling, equipment troubleshooting, SOP compliance, asset documentation
Root Cause Analysis: Structured fault diagnosis, defect elimination, decision analysis
Safety & Compliance: Aviation and industrial safety standards, regulatory compliance, risk-aware operations
Software & Reporting: MS Excel, Word, PowerPoint — technical reporting and maintenance documentation

KEY ENGINEERING PROJECTS

- Water Jar Internal Cleaning System** — Designed and fabricated a motor-driven internal water jar cleaning system (shaft, bearing selection, brush, mechanics); FEA validation under operational loads.
- Composite Motorbike Helmet** — Material selection, SolidWorks modelling, Ansys impact simulation, hand layup fabrication, standard helmet impact testing comparison with the simulation results— full design cycle.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Engineers Australia — Professional Mechanical Engineer (2026)**
- Fundamental Skills in Engineering Design, University of Leeds (2023)
 - Material Selection for Mechanical Design, Udemy (2024)